



Deep reinforcement learning-driven intelligent portfolio management with green computing: Sustainable portfolio optimization and management

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ABSTRACT

Portfolio management remains a key area in quantitative trading. To address limitations in existing deep reinforcement learning (DRL)-based models, which fail to adapt trading strategies and properly utilize supervisory information, we propose a Dynamic Predictor Selection-based Deep Reinforcement Learning (DPDRL) model. The DPDRL model integrates multiple predictors to forecast stock movements and dynamically selects the most accurate predictions, optimizing investment allocation via a market environment evaluation module. Our model was evaluated using daily candlestick data from the SSE 50 and CSI 500 indices. The results show that DPDRL outperforms other models in key evaluation metrics: it achieves a 48.99% Annualized Rate of Return (ARR), a Sharpe ratio of 2.34, an Annualized Volatility (AVoL) of 0.1390, and a Maximum Drawdown (MDD) of 8.21%, significantly improving risk-return performance. Ablation experiments confirm the contributions of the dynamic predictor selector and market evaluation module to the model's accuracy and decision-making quality.

1. Introduction

In recent years, with the comprehensive deepening of reforms and the development of economic policies, China's market economy has grown rapidly, and the global economy has entered a new stage [1]. The international financial market is showing a trend of diversification and integration. In this context, the demand for wealth appreciation among enterprises and individuals has become increasingly urgent, manifested in the increasing variety and widespread attention of financial products. As people's understanding of financial management and investment deepens, they can choose from more financial products to achieve higher asset appreciation and returns. At present, this situation is becoming increasingly common, with stock and securities portfolio management being particularly common [2].

An investment portfolio is essentially a collection of multiple assets, which can include stocks, bonds, futures, options, and so on [3]. The process of constructing an investment portfolio refers to buying or selling various assets based on a certain proportion on a fixed principal basis, and continuously adjusting these proportions. By diversifying investments, it is possible to reduce holding risks and pursue the highest possible returns. In financial markets, the standard deviation of portfolio value is commonly used to measure risk, while the portfolio's profit margin is used to represent returns. For individual investors, optimizing

asset allocation to achieve maximum long-term returns is very difficult. Even investment experts from financial institutions need to invest a lot of time and effort, applying professional knowledge to analyze each asset comprehensively in a noisy and unstable market. Therefore, traditional manual analysis is not only complex, but also costly [4].

With the continuous advancement of information technology, deep reinforcement learning (DRL) has shown great potential in the financial field [5]. The core idea of this method is to train and optimize the model through the interaction between the agent and the environment, which not only saves time in obtaining annotated samples, but also endows the model with the ability to learn autonomously. However, existing methods still face many unresolved issues. On the one hand, most methods only focus on a single prediction problem and fail to fully consider the cyclical characteristics of the stock market. Agents often only learn one market forecasting method, which may become unsuitable during rapid market changes. When there are significant differences in the market, agents often need to learn new prediction methods at the cost of forgetting old ones, making it difficult to find suitable strategies in volatile situations. On the other hand, some investment strategies do not consider the proportion of funds, that is, they do not take into account the distribution ratio of the total investment amount among different stocks. Although deep learning can effectively predict the future trends of stocks and rank them reasonably based on market

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cyclical changes, it still lacks sufficient consideration in terms of the proportion of capital investment [6].

Considering the changes in investment strategies and economic cycles, this paper proposes a stock prediction method that takes into account different trading patterns to enhance the trading decision-making ability of the model. Agents learn appropriate trading strategies in different market environments at different times, improve the accuracy of predicting different assets, and optimize the allocation of investment funds. Most methods predict stock trends based on historical data, but different investment experts have different insights into the market. This article constructs an asset weight module to enable agents to learn different market interpretation methods to train multimodal predictors, while extracting time series features and combining different predictors to select the best prediction results in a reinforcement learning (RL) environment. In addition, considering the correlation between individual stock trends and market trends, this article also constructs a market environment evaluation module to dynamically adjust the proportion of funds. The main contributions of this article are as follows:

(1) In order to address the problem of single prediction, this paper designs a stock prediction method suitable for multiple trading models in RL networks. Agents will learn these trading patterns and choose the most suitable trading pattern based on the current market environment, thereby solving the problem of poor performance of agents in different time periods caused by a single prediction pattern.

(2) In order to solve the problem of capital ratio, this article divides the investment portfolio problem into two parts: investment weight problem and investment ratio problem, and designs two different modules to deal with these two problems, namely the asset weight module and the market environment evaluation module. The market environment evaluation module dynamically adjusts the capital ratio by analyzing market data to optimize the investment portfolio.

The remaining organization of this article is as follows: 2 provides a comprehensive review of related works, highlighting the key contributions and limitations of previous portfolio management models, including those based on DRL. 3 presents the methodology, where we introduce our DPDL model, explaining its components and how they address the limitations identified in existing models. 4 discusses the experimental setup and presents the results of our model, including comparisons with existing methods and performance metrics. Finally, 5 concludes the paper by summarizing the findings, discussing the implications of the results, and outlining directions for future work.

2. Related works

Currently, research on investment portfolios remains a hot topic both domestically and internationally. This section mainly introduces the research status of traditional investment portfolio strategies, investment portfolio strategies based on time series prediction, and investment portfolio strategies based on DRL, and summarizes them

The traditional portfolio management analysis requires evaluating the returns and risks of each asset, but its results often lack credibility and stability due to subjective factors. Markowitz[7] proposed the Mean Variance (MV) model, pioneering modern portfolio theory. Subsequently, research mainly focused on the following branches: multi-objective problems, practical factor handling, heuristic methods, and the application of fuzzy theory. In the process of building investment portfolios, researchers often need to consider multiple indicators simultaneously, such as returns, risks, and transaction fees [8]. In the process of portfolio selection, investors often have personal behavioral preferences, which may be vague, uncertain, and incomplete. Guo et al. [9] first combined fuzzy preference relationships (FPRs) with portfolio theory to study investor choice problems. He et al. [10] considered the subjectivity and differences of investors when investing in the securities market, proposed a combination evaluation system based on variable index hesitant fuzzy language set, obtained the optimal strategy by solving the equivalent nonlinear model, and established an optimized

group decision-making combination strategy to improve returns or reduce risks. Lin et al. [11] proposed a modified Sharpe ratio to construct the optimal investment portfolio under systematic events. When there is no short selling restriction, solve the allocation problem through analytical methods; When there are short selling restrictions, solve them through numerical methods. Compared with the mean variance, equal weight, and global minimum variance combinations, the designed combination performs better.

Time series prediction is a method that utilizes a large amount of past behavior time series data to make long-term or short-term predictions, and has wide applications in many fields [12]. The commonly used prediction methods are divided into statistical methods and machine learning methods. Statistical methods include autoregressive integrated moving average and exponential smoothed state space models [13,14]. In stock trend prediction, the performance of these methods in financial markets is often affected due to the possibility of drastic fluctuations in stock prices in a very short period of time.

Machine learning based prediction schemes have been used in stock prediction and have been proven to be able to capture complex patterns in heterogeneous data using powerful nonlinear models. Behera et al. [15] proposed a hybrid approach that includes a machine learning algorithm for predicting stock returns and a mean value at risk model for portfolio selection, which is a unique portfolio building technique to achieve effective investment strategies. In reference [16], the author designed a portfolio consisting of weighted investment portfolio, global minimum variance investment portfolio, market value weighted investment portfolio, and maximum Sharpe ratio investment portfolio. The results showed that all three optimized investment portfolios outperformed the Nifty index. Zhao et al. [17] demonstrated how to apply machine learning algorithms to distinguish between high-quality stocks and low-quality stocks. They utilized different deep learning algorithms to construct 244 techniques and basic features to describe each stock, and labeled the stocks based on their ranking of stock return volatility. The performance of these deep learning algorithms is superior to traditional algorithms.

The RL finds the optimal strategy for controlling actions through the continuous interaction between agents and the environment. Currently, some researchers are combining deep neural networks with RL to learn trading strategies. Huang et al. [18] proposed a DRL framework for financial signal processing and online portfolio optimization, which utilizes online gradient descent to quickly converge and solve the DRL algorithm, while extending fuzzy learning to reduce uncertainty in financial time series. Liu et al. [19] combined deep learning with reinforcement learning (RL) to propose a multi-source fusion DRL (MSF-DRL) framework. This framework integrates stock data, technical indicators, and candlestick charts to address the variations in different types of data in complex stock market environments. Kwon et al. [20] proposed a new hybrid decision model that combines rule-based systems with DRL. This model is designed to adapt to different market scenarios, withstand market pressure, reduce trading costs, and enhance scalability. Massahi et al. [21] addressed the issue of sparse trading data by proposing a method that combines Q-learning (QL) with transfer learning. By transferring pre-trained models, this approach reduces the iterative search space and is applied to portfolio optimization. Li et al. [22] proposed a dynamic DRL portfolio model that can adapt to the introduction of new assets without requiring additional training. The advantages and limitations of the above methods are summarized in Table 1.

However, these methods train only one investment strategy, making it difficult to capture specific characteristics amid market changes. To address this issue, this paper designs a stock prediction method under multiple trading patterns, where the agent learns these patterns and selects the most appropriate one for the current situation.

Table 1
Summary of method characteristics.

Reference	Contributions	Limitations
[8]	This paper proposes a multi-task learning framework for optimizing stock portfolios, focusing on both risk and return. It enhances traditional portfolio optimization by incorporating multiple tasks to handle various factors affecting financial decision-making.	The method may face challenges in scalability and computational efficiency when applied to large datasets or real-time trading environments.
[9]	The paper introduces fuzzy preference relations to handle the uncertainty and subjectivity in investment decisions. It provides a novel way to model portfolio selection with subjective criteria from decision-makers.	The use of fuzzy logic might lead to overcomplicated models that are difficult to implement in real-world applications without simplification.
[10]	The paper improves portfolio optimization by considering multiple decision-making criteria through a fuzzy linguistic approach, integrating group decisions for asset allocation.	The method may not generalize well to markets with extreme volatility or sudden shifts, as it relies on historical decision-making patterns.
[11]	This research focuses on portfolio selection while accounting for systemic risks that affect entire financial systems, improving robustness.	The model may not be adaptable to dynamic market conditions where systemic risks evolve rapidly.
[12]	While not focused on portfolio management, this paper applies deep learning, particularly LSTM networks, for short-term traffic prediction, which could be adapted to predict market behavior.	The approach is specialized for traffic data and may require significant adjustments to apply to financial markets with different characteristics.
[13]	This paper develops a dynamic trading model combining gray models and ARIMA for improved time series forecasting, which is crucial in financial decision-making.	The model may struggle with non-linear market conditions, as traditional models like ARIMA are more suited for linear trends.
[14]	The paper improves financial time series prediction by using nonlinear exponential autoregressive models, enhancing prediction accuracy.	The model might be computationally expensive and may not scale well for large financial datasets.
[15]	This paper integrates machine learning with mean-value-at-risk for portfolio optimization, improving risk management.	It relies on historical data and may not perform well during market crashes or extreme events.
[16]	The paper focuses on real-time portfolio management using machine learning techniques, allowing for quick adaptation to market changes.	Real-time processing requires high computational power and might not be feasible for all types of investors.
[17]	This work uses deep learning to predict stock price movements, offering an integrated framework for more accurate forecasting.	Deep learning models can overfit to past data and might not generalize well during market shifts.
[18]	This paper proposes a multi-agent reinforcement learning framework for optimizing trading strategies, enhancing adaptability in real-world trading scenarios.	Multi-agent frameworks can be complex and require significant computational resources to train effectively.
[19]	The paper integrates various data types with deep reinforcement learning for algorithmic trading, enhancing predictive power.	Integrating multiple data types may lead to overfitting if not carefully managed.
[20]	This research combines rule-based systems with deep	The hybrid model may introduce complexity in decision-making,

Table 1 (continued)

Reference	Contributions	Limitations
	reinforcement learning to develop adaptive trading strategies.	which could slow down real-time execution.
[21]	This paper proposes a deep Q-learning model for algorithmic trading in commodity futures markets, improving trading decisions.	The model may not adapt well to markets with little historical data or extreme market fluctuations.
[22]	This paper utilizes high-frequency data and deep reinforcement learning for online portfolio management, making real-time adjustments to maximize returns.	Handling high-frequency data can be computationally intensive and may require advanced data management techniques.

3. Methodology

3.1. Problem description

The task of portfolio management is to maximize investment returns while assuming certain risks, and investment managers need to review the timing to change the weights of each asset class. For example, if an investment manager determines that the overall condition of a certain asset is more favorable to investors compared to other assets at the next moment, they need to transfer the weights of the investment portfolio to that asset. Here, the investment manager is our deep reinforcement learning agent. I use stocks as risk assets and treasury bond bonds as risk-free assets to form the asset package of the portfolio.

Stock trading data is a common financial time series, which refers to financial data indexed by time (minutes or days). This article takes days as the trading cycle and represents the stock time series as follows:

$$X = (x_1, x_2, \dots, x_t, \dots, x_n) \quad (1)$$

where X represents the time series of stocks with a trading period of days, and x_k represents the eigenvector of the t -th trading day. When using raw stock trading data as feature vectors:

$$\hat{x}_t = (x_t^o, x_t^c, x_t^h, x_t^l, x_t^v)^T \quad (2)$$

where, x_t^o , x_t^c , x_t^h , x_t^l , and x_t^v represent the opening price, closing price, highest price, lowest price, and trading volume within the first trading day, respectively.

3.2. Dynamic predictor selection based deep reinforcement learning model

This section proposes a dynamic predictor selection based deep reinforcement learning model (DSDRL), with Sharpe ratio as the optimization objective of the algorithm. The overall framework of the algorithm is shown in Fig. The method designed in this article includes multiple parts. Firstly, the asset data x_t is input into the LSTM layer, and attention mechanism is used to recalculate the hidden layer as input for t predictors, thereby obtaining the predicted score y_t of the future stock. At the same time, the recalculated hidden layer is combined with predictive loss memory as input to the selector to generate classification scores. The selector dynamically selects the most suitable predicted score based on the classification score, which serves as the actual stock rating v . Next, input market data $\hat{x}_t$ and convert it into Gaussian distribution parameters through a fully connected layer. Finally, through portfolio management, the stock rating v is converted into a short ratio b and a long ratio w , and the proportion of long funds is determined by the parameter ρ . The entire process is optimized through reinforcement learning.

3.2.1. Markov decision process

Portfolio management is a sequential decision-making process that allocates resources into a group of financial assets based on current

market conditions, with the aim of maximizing returns while controlling risks. This process conforms to the Markov decision process (MDP), in RL, MDP is usually defined as a quadruple $M = \langle S, A, P, R \rangle$.

- (1) $S = (s_1, s_2, \dots, s_t, \dots)$ is the set of all environmental states. In portfolio management problems, the environment includes all available assets in the market and the expectations of all market participants for these assets. In this article, I use the feature vectors of each asset x_t^c and x_t^h as inputs. To simplify the problem, I use the method of looking forward a window w on trading day t to represent the environmental state s . The data preprocessing method is to divide the data in each window of the training set, validation set, and testing set by the maximum value of the offer price within that window, that is:

$$s_t = \begin{bmatrix} x_{t-w+1}^c/x_{max}^c & x_{t-w+1}^h/x_{max}^h \\ x_{t-w+2}^c/x_{max}^c & x_{t-w+2}^h/x_{max}^h \\ \vdots & \vdots \\ x_t^c/x_{max}^c & x_t^h/x_{max}^h \end{bmatrix} \quad (3)$$

where x_{max}^c is the maximum closing price of the asset on the corresponding set.

- (2) $A = (a_1, a_2, \dots, a_t, \dots)$ is the action space, and the action $a_t = [v, \rho]$ executed at time t . v represents the rating of different stocks by the agent, and based on the rating, selects stocks suitable for long positions for investment, and calculates the investment ratio $w_t \in R^{n \times T}$, where n is the number of selected stocks, and p is the proportion of long funds.
- (3) R is the reward function, which is the cumulative reward obtained by the reinforcement learning agent after executing one round, and r_t is the immediate reward obtained after executing the action in state s_t . In the single asset problem, r_t has two types: simple rate of return and logarithmic rate of return, which are expressed as:

$$r_t^{simple} = \frac{x_t^c - x_{t-1}^c}{x_{t-1}^c} \quad (4)$$

$$r_t^{ln} = \ln\left(\frac{x_t^c}{x_{t-1}^c}\right) \quad (5)$$

Let the volatility vector of the investment portfolio price be y_t :

$$y_t = x_t^c / x_{t-1}^c = \left(1, \frac{x_{t-1,1}^c}{x_{t-1,1}^c}, \frac{x_{t-1,2}^c}{x_{t-1,2}^c}, \dots, \frac{x_{t-1,m}^c}{x_{t-1,m}^c}\right)^T \quad (6)$$

Then immediately report r_t as:

$$r_t = \ln(y_t^T \cdot a_t) \quad (7)$$

The cumulative return R is

$$R = \sum_{t=0}^T r_t \quad (8)$$

Let the initial portfolio value is p_0 , then the portfolio value p_t at the end of day t is

$$p_t = p_0 e^{\sum_{i=0}^{t-1} r_i} = p_0 \prod_{i=0}^{t-1} y_i^T \cdot a_i \quad (9)$$

P is the probability of state transition, where $p(s_{t+1}|s_t, a_t)$ is defined as the probability of taking action a at transitioning from state s_t to state s_{t+1} . I assume that MDP satisfies $p(s_{t+1}|s_t, a_t, \dots, s_1, a_1) = p(s_{t+1}|s_t, a_t)$. Since the complex market environment, I cannot know the probability of transitioning from one state to another, but model free reinforcement learning methods overcome this drawback by using sampling methods.

3.2.2. Transaction process

In the real process of stock trading, buying and selling stocks are not free, and transaction fees consist of stamp duty, transfer fees, and brokerage commissions. Due to changes in stock prices, the weights of investment portfolios may shift. Assuming that at the beginning of day t , the weight vector is a_{t-1} , it will transition to a'_t at the end of the day:

$$a'_t = \frac{a_t - a_{t-1}}{a_{t-1}} \quad (10)$$

where $*$ represents element wise multiplication. At the end of day t , the investment manager changed a'_t to a_t through buying and selling operations, resulting in transaction costs, as shown in Fig. 1.

3.3. Asset weight module

The asset weight module is a strategy $\pi_{\theta_1}^t(v|x_t; \theta_1)$, and its input is asset data x_t . Output as asset rating v . Firstly, long short-term memory layers are used to extract the temporal features of assets, and then attention mechanisms are used to effectively model over a long time span. The obtained information is passed on to the predictor to score the stocks. Part of the results obtained from the predictor are passed on to the asset management module to obtain the invested stocks and corresponding weights, while the other part is passed on to the reinforcement learning environment and stored in the loss memory.

3.3.1. Asset Scoring

First, LSTM is used to extract features from the input asset $X = (x_1, x_2, \dots, x_t, \dots, x_n)$, with the calculation formula as follows:

$$h_{t-i+1} = LSTM(h_{t-i}, x_t), i = 0, 1, 2, \dots, t - T + 1 \quad (11)$$

where h_{t-i+1} represents the hidden state encoded by LSTM at time t , and C is the dimension of the hidden layer. The last hidden state h_t can be seen as a global representation of the input signal. In order to effectively model nonlinear relationships over a long time span, a time attention mechanism is adopted to adaptively model nonlinear relationships. The attention weight calculation formula is:

$$e_t = V_e^T \tanh(W[h_T; h_t] + Ux_t^m) \quad (12)$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{i=1}^K \exp(e_i)} \quad (13)$$

where $V \in R^C$, $W \in R^{C \times 2C}$, and $U \in R^{C \times C}$ represent the learning parameters, respectively. We recalculate the final hidden state as:

$$\hat{h}_t^a = \sum_{i=t-T+1}^t \alpha_i h_i \quad (14)$$

Then, the hidden state $\hat{h}_t^a \in R^{C \times m}$ obtained is combined with the loss memory L_t as the input for *actor*, as shown below:

$$v = \text{actor}(\hat{h}_t^a, L_t) \quad (15)$$

The actor layer consists of a selector and an predictor, respectively.

3.3.2. Predictor and selector

A) Predictor: The predictor consists of m networks, each of which outputs a predicted result, simulating the decision results of m experts, given by the following formula:

$$y_t^i = \text{actor}_i(\hat{h}_t^a), i = 1, 2, \dots, m \quad (16)$$

$$y_t = (y_t^1, y_t^2, \dots, y_t^m) \quad (17)$$

where actor_i adopts a fully connected layer. Transfer the prediction result of the t -th time predictor to the loss function, and save the

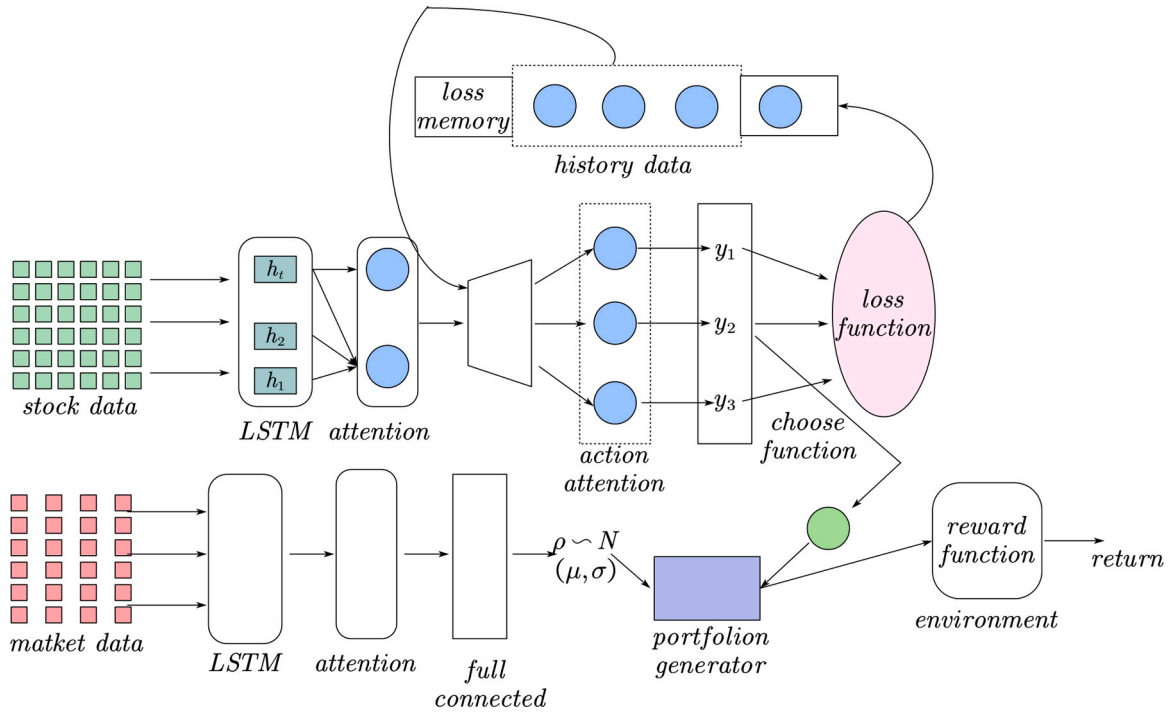


Fig. 1. Framework diagram of DSDRL algorithm.

calculated result to the loss memory L . The calculation formula is as follows:

$$l_t^i = \text{loss}(y_k^i), i = 1, 2, \dots, m \quad (18)$$

$$l_k = [l_k^1, l_k^2, \dots, l_k^m] \quad (19)$$

$$L \leftarrow [L, l_k] \quad (20)$$

The loss function will calculate all actions y_m . The difference between the same report and the true same report does not participate in network updates. It outputs the results of the network and interacts with the reinforcement learning environment to calculate the prediction loss, which is stored in the prediction loss memory.

B) Selector: A selector is a dynamic selector that combines historical predictive memory loss with current market state information to select the most suitable investment expert at the current time. Its calculation formula is as follows:

$$p_1, p_2, \dots, p_m = \text{Routing}(\hat{h}_t, L_t) \quad (21)$$

$$L_t = [l_{t-1}, l_{t-2}, \dots, l_{t-k}] \quad (22)$$

where m is the number of classifier classifications, Routing is the fully connected layer, p_m is the classification score, and L_t is the memory loss from time t to time $t-1$. Finally, based on p_i , select the most suitable asset rating v , and the calculation formula is as follows:

$$z = \arg \max(p_i), i = 1, 2, \dots, m \quad (23)$$

$$v = y_t^z$$

During the training process, at the beginning of each iteration, the loss memory needs to be cleared and filled with a zero vector of size m . When a new loss enters the loss memory, the oldest memory loss will be eliminated. The entire process is shown in Fig. 2.

3.4. Market evaluation module, portfolio strategy and optimization

3.4.1. Market evaluation module

In previous reinforcement learning-based investment models, the

final investment strategy only analyzes individual stocks while neglecting market changes, resulting in the allocation of all funds to stock investments even when market conditions deteriorate.

The market environment evaluation is a strategy function $\pi_{\theta_2}(v|\hat{x}, \theta_1)$, where the input is market data $\hat{x}_t$ and the output is the holding capital proportion ρ . For market data $\hat{x}_t$, the LSTM layer extracts hidden states, which are then processed by a temporal attention mechanism to capture information over a longer period $\hat{h}_t^b \in R^{C \times m}$. Subsequently, a fully connected layer computes μ and σ .

3.4.2. Portfolio strategy

After calculating the asset score u , use the softmax function to convert the score v into weights for the investment portfolio. Rank the scores obtained from assets, select the top ranked stocks for long positions, and classify them as set G ; Select stocks with lower rankings for short selling and classify them as set H . The ratio of long to short is as follows:

$$w_i = \begin{cases} \frac{\exp(v_i)}{\sum_{j \in G} \exp(v_j)}, & i \in G \\ 0, & i \notin G \end{cases} \quad (24)$$

$$b_i = \begin{cases} \frac{\exp(v_i)}{\sum_{j \in H} \exp(v_j)}, & i \in H \\ 0, & i \notin H \end{cases} \quad (25)$$

3.4.3. Strategy optimization

Optimize the investment portfolio strategy $\pi(X, \hat{x}_t, \theta)$. The strategy π is divided into two parts:

- 1) By inputting stock data $\pi_{\theta_1}(v|X, \theta_1)$, the proportion of short and long stocks can be obtained, and the return for holding period t is $r_t = \frac{p_t}{p_{t-1}} - \pi_{\theta}$. For the trajectory τ of RL, the calculation formula for achieving the maximum value of the target by optimizing the network parameter θ_1 is as follows:

$$\theta^* = \operatorname{argmax}_{\theta_1} J^a(\theta) = \operatorname{argmax}_{\theta_1} \sum_{\tau=1}^T \prod_{t=1}^{\tau} \frac{p_t}{p_{t-1}} \pi_{\theta_1} \quad (26)$$

$$J(\theta) = J^a(\theta_1) + \omega J^b(\theta_2) \quad (28)$$

2) Obtain the proportion of long funds by inputting stock data $\pi_{\theta_2}(v|\hat{x}, \theta_1)$. For the trajectory τ of RL, given a reward $R(\tau)$, the calculation formula for optimizing the network parameter θ_2 is as follows:

Algorithm 1. Describes the entire process of the reinforcement learning algorithm based on the dynamic selection predictor.

Algorithm 1. RL algorithm based on dynamic selection predictor

Input: X and $\hat{x}$

Output: θ_1 and θ_2

Step 1: Initialize loss memory M and experience replay D . Initialize behavior network θ_1

(stock rating) and θ_2 (investment amount)

Step 2: for $i \leftarrow 1$ to m do

 Initialize X and $\hat{x}$

 Preprocessing status

$$h_1^a = \text{LSTM}(X^a), h_1^b = \text{LSTM}(X^b), \mu, \sigma = \text{FC}(h_1^b)$$

 for Every time the environment is updated do

$$y_t = \text{actor}(h_t^a), l_t = \text{loss}(y_t), M = [M, l_t]$$

$$v \leftarrow \pi_{\theta_1}(v | X)$$

$$\rho \leftarrow \pi_{\theta_2}(\rho | \hat{x})$$

$$\text{Update the next state } s_{t+1} \sim \rho(s_{t+1} | s_t, \rho, v), s_t = [X_t, \hat{x}_t]$$

$$\text{Update experience replay } D \leftarrow D \cup \{(s_t, \rho, v, r(s_t, \rho, v), s_{t+1})\}$$

 End for

Step 3: for $j \leftarrow 1$ to n do

 Collect data from experience replay and update network parameters $\theta \leftarrow \theta + \eta \nabla J(\theta)$ using equation (28).

 End for

 End for

$$\theta^* = \operatorname{argmax}_{\theta_2} J^b(\theta) = \operatorname{argmax}_{\theta_2} \sum_{\tau} R(\tau) \cdot \pi_{\theta_2}^b(\tau) \quad (27)$$

Combining (26) and (27) for weighted summation to obtain the final daily scale function:

4. Experimental results

4.1. Experimental preparation

This section first introduces data processing, baselines, and measurement standards; Then evaluate the algorithm of this article using

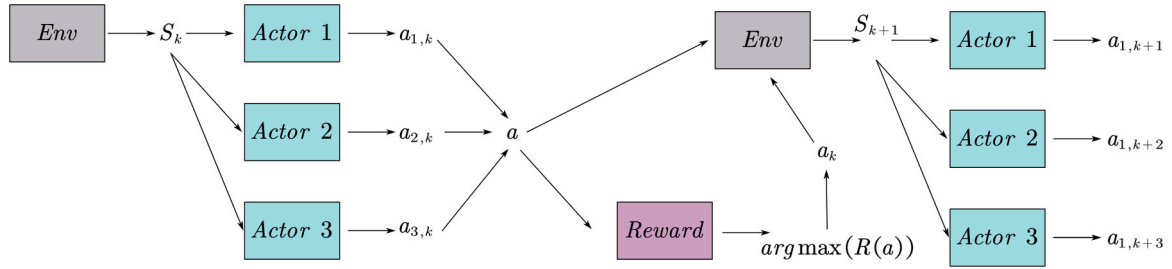


Fig. 2. Track diagram of reinforcement learning.

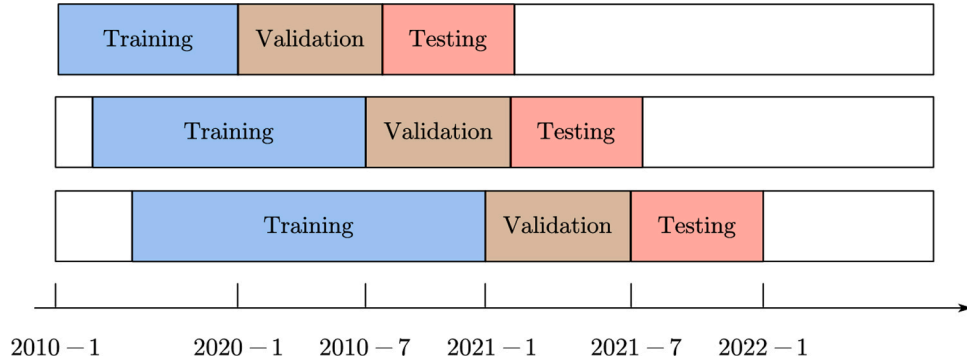


Fig. 3. Dataset segmentation of sliding training.

baselines in different markets; Finally, through ablation experiments, the impact of the dynamic selection predictor in the asset weight module and the market environment evaluation module on rewards was analyzed. The experiment is based on PyTorch 2 under Linux system 0.1 framework, hardware configuration is Intel Xeon CPU E5-2697R v4 @ 2.30 GHz, GPU RTX 3080.

4.2. Experimental datasets

4.2.1. Datasets

This article collects constituent stocks of the SSE 50 Index and the CSI 500 Index from the Tushare (<https://tushare.pro/>) database as datasets. Due to the semi annual adjustment of the index's constituent stocks, some new stocks need to be excluded, resulting in the collection of 40

and 382 constituent stocks respectively. This article selects the SSE 50 Index and CSI 500 Index as the corresponding datasets for the overall market data. Table 1 shows the fields of experimental data. To ensure the proportional constraints on portfolio weights, we acknowledge the limitation of using a penalty term in the reward function to enforce the sum-to-zero constraint on the weight vector. To address this, we have implemented a hard constraint approach using a projection method. After each action update, the weight vector is projected onto the feasible set to ensure the constraint is strictly satisfied at all times. This projection is computationally efficient and maintains the interpretability of the reinforcement learning model.

4.2.2. Training set settings

For the model, some hyperparameters are set as follows: the holding period is 30 days, including 29 days of holding and 1 day of trading. The learning rate for reinforcement learning is set to 1×10^{-5} , and the discount factor γ is set to 0.96. To more realistically simulate investors' strategies, the number of prediction classifiers is set to 3. All weights are pre-processed and initialized, and the LSTM output space size is set to 128 with each input data batch size being 128. The ReLU activation function and Adam optimizer are used, with the maximum number of iterations set to 500. The initial capital is 100,000 yuan, and the transaction fee rate is 0.1 %. To better capture sequential patterns in stock data, a rolling training, validation, and testing method is adopted. The training process is shown in Fig. 4, and the specific steps are as follows:

1) Training: Select samples over a period (in trading days) and train over multiple periods. 2) Validation: Select samples after the training window for validation. If the validation loss does not significantly decrease over multiple periods, stop training. 3) Testing: Select samples after the validation period for testing. 4) Forward rolling window: After testing, add the validation samples to the training dataset and roll the window forward. 5) Repeat: Repeat the above training process.

In the experiment, grid search is used to adjust the lengths of the training, validation, and testing periods, as shown in Fig. 3. The specific settings are as follows: Training window lengths: {1200, 1500, 1800, 2100, 2400}. Validation window lengths: {120, 150, 180, 210, 240}.

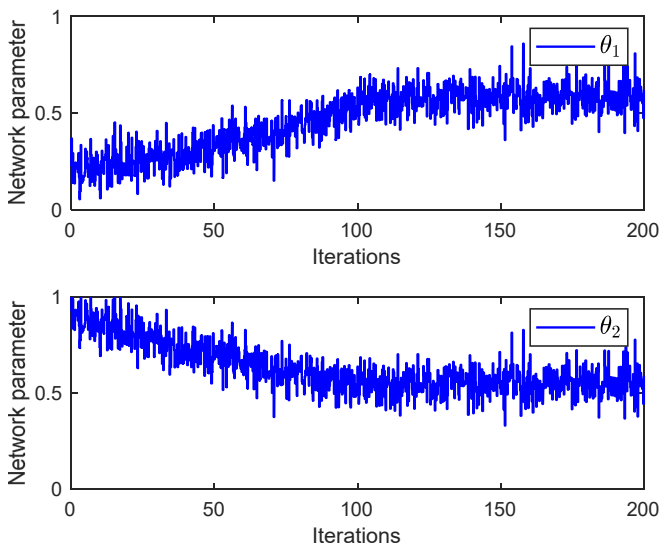


Fig. 4. The training process of network parameters under the algorithm.

Testing window lengths: {120, 150, 180, 210, 240}.

4.3. Evaluation metrics and comparison methods

4.3.1. Evaluation metrics

In the experiment, five metrics are used, categorized into three types:
Profit Metrics: Annualized Rate of Return (ARR).

Risk Metrics: Annualized Volatility (AVOL) and Maximum Draw-down (MDD).

Risk-Return Metrics: Annualized Sharpe Ratio (ASR) and the Risk-Return Ratio (CR).

Lower values of AVOL and MDD are preferred, while higher values of the remaining metrics are desirable.

4.3.2. Comparison methods

To effectively evaluate the deep reinforcement learning portfolio model based on dynamic selectors proposed in this paper, the following five methods are compared:

FinRL: A deep RL library for quantitative finance, selecting the DDPG and PPO methods, and using the best-performing method.

TRA: Samples are allocated to different predictors based on their belonging to different time periods, resulting in dynamic scoring for different stocks to implement portfolio strategies.

DT: Composed of market scoring units, stock scoring units, and portfolio managers, this is a reinforcement learning-based portfolio method.

DPRL: A model-based dynamic portfolio optimization deep RL method, trained using a framework with prediction modules, adversarial data augmentation modules, and behavior cloning modules.

DSDRL: Uses a dynamic selection module to choose the most suitable scoring strategy for the current time period, combined with a market environment evaluation module to derive the portfolio strategy.

4.4. Experimental results

4.4.1. Algorithm performance

Since the characteristics of financial markets, many reinforcement learning methods used for portfolio problems exhibit significant fluctuations during training, making it difficult to converge, especially for reinforcement learning algorithms targeting individual stock investments. The performance of the proposed method during the iteration process of network parameters is shown in Fig. 3, and the return rate is shown in Fig. 4 and Fig. 5. As can be seen from Fig. 4, the proposed method shows a steady overall improvement during the training process, reaching its peak at 100 iterations, without experiencing severe fluctuations in subsequent training.

4.5. Comparison with baseline methods

From Tables 2 and 3, it can be seen that among the methods based on

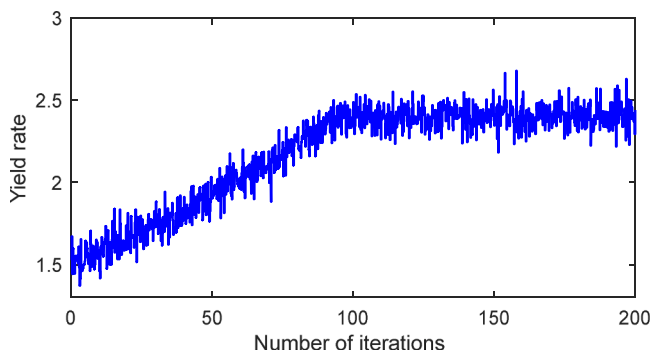


Fig. 5. Algorithm training performance.

Table 2

Experimental data fields.

Field Name	Explain
Close	Closing price
Volume	Turnover
OBV	Energy Tide Index
MACD	Smooth moving average indicator
WPR	Williams percentage interval indicator
TSI	Real intensity index indicator
AO	Awesome Oscillator
RSI	Relative strength index
VWMA	Volume weighted moving average indicator

Table 3

Backtesting performance of all methods in SSE 50.

Methods	ARR(%)	ASR	AVoL	MDD(%)	CR
FinRL	9.43	0.47	0.3690	22.12	1.2689
TRA	14.34	0.81	0.4633	23.12	2.2144
DT	22.68	1.28	0.2911	19.28	1.4588
DPRL	36.37	0.89	0.2833	18.22	1.5622
DSDRL	48.99	2.34	0.1390	8.21	3.5619

deep RL, the DSDRL method performs better. By comparing other evaluation metrics, it is evident that DSDRL achieves higher cumulative returns on both datasets compared to other methods, and the Sharpe ratio has significantly improved. This means that investors can achieve higher returns for the same unit of risk. In terms of risk assessment metrics (AVOL and MDD), the performance of DSDRL is generally superior to other methods, further confirming the effectiveness and robustness of this strategy.

Its superior risk control performance arises from the dynamic selection of predictors and the market environment evaluation module, which together allow the model to adaptively allocate portfolio weights and reduce exposure to risky assets in volatile market conditions, effectively managing risk through implicit adjustments in response to market fluctuations.

Table 4 shows that the evaluation results of the SDE-RF algorithm achieved notable F1-score when dealing with imbalanced datasets, indicating the efficiency of the proposed algorithm in this paper. (Table 5)

Table 4

Backtesting performance of all methods in CSI 500.

Methods	ARR(%)	ASR	AVoL	MDD(%)	CR
FinRL	8.23	0.80	0.5422	32.23	1.2456
TRA	12.14	0.76	0.4834	26.33	2.4565
DT	24.33	1.45	0.7022	21.56	1.7643
DPRL	34.79	1.89	0.3611	19.31	1.2356
DSDRL	46.99	3.88	0.1561	7.33	3.6412

Table 5

Comparison of SDE-RF algorithm with three related work.

Methods	Classifier	F1 -score %
FinRL	RF	65.25
TRA	RF	56.87
DPRL	Ensemble of oversampling and under sampling approach	78.33
DSDRL	Optimized RF (SDE-RF algorithm)	3.88

Table 6

Backtest performance of ablation test.

Methods	ARR(%)	ASR	AVoL	MDD(%)	CR
DSDRL-1	29.33	1.42	0.301	21.78	1.292
DSDRL-2	36.79	1.39	0.262	19.23	1.591
DSDRL	48.99	2.34	0.139	8.21	3.562

4.6. Ablation experiments

To fully explain the importance of the dynamic selector predictor module and the market environment evaluation module, ablation experiments were conducted on the same dataset for both parts. The analysis of the two ablation experiments is as follows:

1. Ablation of the Dynamic Selector Predictor: In this study, multiple prediction methods were trained, and the agent selected the optimal prediction results to form the investment portfolio. In this experiment, the dynamic selector predictor module was removed, meaning the model would train only one prediction method, and the agent would form the investment portfolio based on a single prediction result. We denote this ablation experiment as DSDRL-1.
2. Ablation of the Market Environment Evaluation: To allow the agent to consider the impact of the market environment and allocate funds based on market data, this experiment removed the market environment evaluation module, meaning the agent would not consider fund allocation, and all funds would be used for long positions. I denote this ablation experiment as DSDRL-2.

The results of the ablation experiments are shown in Table 4, with results rounded to three significant figures. The results indicate that the DSDRL method's performance was suboptimal in terms of returns and risk mitigation when the dynamic selector predictor module and the market environment evaluation module were removed. Therefore, the introduction of the dynamic selector predictor module and the market environment evaluation module effectively helps the agent make better decisions. (Table 6)

The ablation experiments show that the introduction of the dynamic predictor selector and the market environment evaluation module significantly improves the model's prediction accuracy and investment decision quality. Removing the dynamic predictor selector leads to a significant decline in prediction performance, indicating its crucial role in handling market changes. Similarly, removing the market environment evaluation module results in less adjustment of investment proportions, reducing the performance of the portfolio. Overall, the ablation experiments validate the contribution of these modules to the model's performance.

The dynamic selection of the trading predictor enhances risk management by allowing the model to adapt to changing market conditions in real time. Unlike a static predictor, which may fail under certain market regimes, our approach evaluates multiple predictors and selects the most suitable one based on recent performance metrics and market conditions. This ensures that the model can dynamically shift towards more conservative strategies in volatile markets and capitalize on high-confidence predictions during stable periods. By continuously optimizing the selection process through reinforcement learning, the model reduces exposure to high-risk assets, improves portfolio diversification, and mitigates potential drawdowns, ultimately leading to more stable and resilient investment performance.

The trading agent selects an appropriate predictor through a reinforcement learning-based selection mechanism rather than introducing a separate action space. Specifically, each predictor generates a confidence score based on historical prediction accuracy and market conditions. These scores are then processed using a softmax-based weighted sampling strategy, where predictors with higher confidence scores are more likely to be selected. This selection process is integrated into the agent's decision-making pipeline, allowing it to dynamically adapt its strategy without explicitly expanding the action space. By leveraging this approach, the model efficiently balances exploration and exploitation, ensuring that the best-performing predictor is used while still allowing flexibility to adapt to new market conditions.

5. Conclusions

In this paper, we proposed the Dynamic Predictor Selection-based DPDL model for portfolio management, which incorporates multiple predictors and a market environment evaluation module to improve prediction accuracy and optimize portfolio allocation. Our experiments on stock market data from the SSE 50 and CSI 500 indices demonstrate the effectiveness of our approach. The DPDL model outperforms existing methods in key evaluation metrics, achieving an ARR of 48.99 %, an ASR of 2.34, an AVoL of 0.1390, and an MDD of 8.21 %. These results show significant improvements in both return and risk management when compared to other methods, such as FinRL, TRA, DT, and DPRL.

Future work will focus on incorporating more diverse prediction methods into the dynamic predictor selector and simulating a wider range of trading strategies, enabling the model to make better decisions in more complex trading environments.

CRedit authorship contribution statement

Yi Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical approval

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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